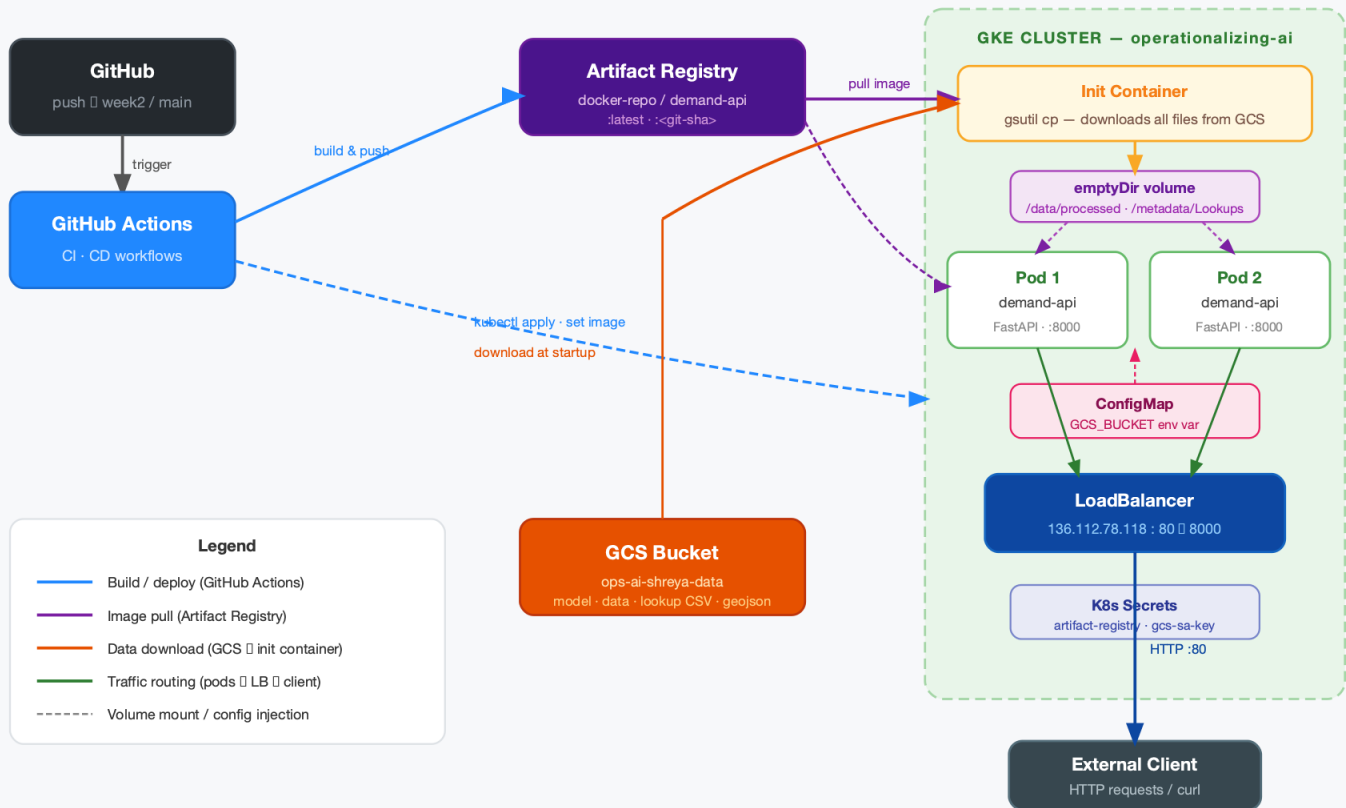


Architecture Diagram

Demand API — Deployment Architecture

GitHub Actions · Artifact Registry · GKE · GCS · ops-ai-shreya



Health Check + Heatmap API + kubectl pods

```
shreyamendi@Mac-1131 Ops-AI-Student % curl http://136.112.78.118/health
{"status":"ok"}
shreyamendi@Mac-1131 Ops-AI-Student % curl "http://136.112.78.118/api/heatmap?hour=17&date=2026-01-15&holiday=regular"
{"demand":{"4":0.68,"13":14.48,"24":4.39,"41":4.05,"43":37.59,"45":2.25,"48":31.8,"50":8.76,"68":46.66,"70":6.18,"74":3.75,"75":10.16,"79":22.7,"87":9.98,"88":5.56,"90":26.4,"93":0.38,"100":24.99,"107":30.79,"113":22.3,"114":15.85,"125":6.34,"132":79.24,"137":19.68,"138":53.96,"140":39.06,"141":41.6,"142":52.6,"143":18.16,"144":15.81,"148":7.93,"151":14.66,"158":13.61,"161":102.8,"162":73.49,"163":58.01,"164":38.47,"166":8.53,"170":52.42,"186":58.02,"195":0.07,"209":2.73,"211":14.71,"229":29.97,"230":53.61,"231":22.17,"233":22.31,"234":49.26,"236":83.25,"237":93.66,"238":33.95,"239":50.91,"246":35.66,"249":23.56,"261":9.96,"262":22.68,"263":32.04},"max":102.8,"hour":17,"dow":4,"holiday":"regular","date":"2026-01-15","is_forecast":false,"data_type":"actual","reference_date":"2026-02-14"}}
shreyamendi@Mac-1131 Ops-AI-Student % kubectl get pods && kubectl get svc demand-api
NAME                                READY   STATUS    RESTARTS   AGE
demand-api-7565995dbc-zf6sr         1/1     Running   0           16m
demand-api-867c968dd5-b9f5w         0/1     Error     5 (2m41s ago)  8m58s
demand-api-867c968dd5-ntnns         0/1     CrashLoopBackOff  5 (57s ago)    8m58s
NAME                                TYPE                                CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
demand-api-LoadBalancer             LoadBalancer           34.118.228.252  136.112.78.118  80:30829/TCP     26m
shreyamendi@Mac-1131 Ops-AI-Student %
```

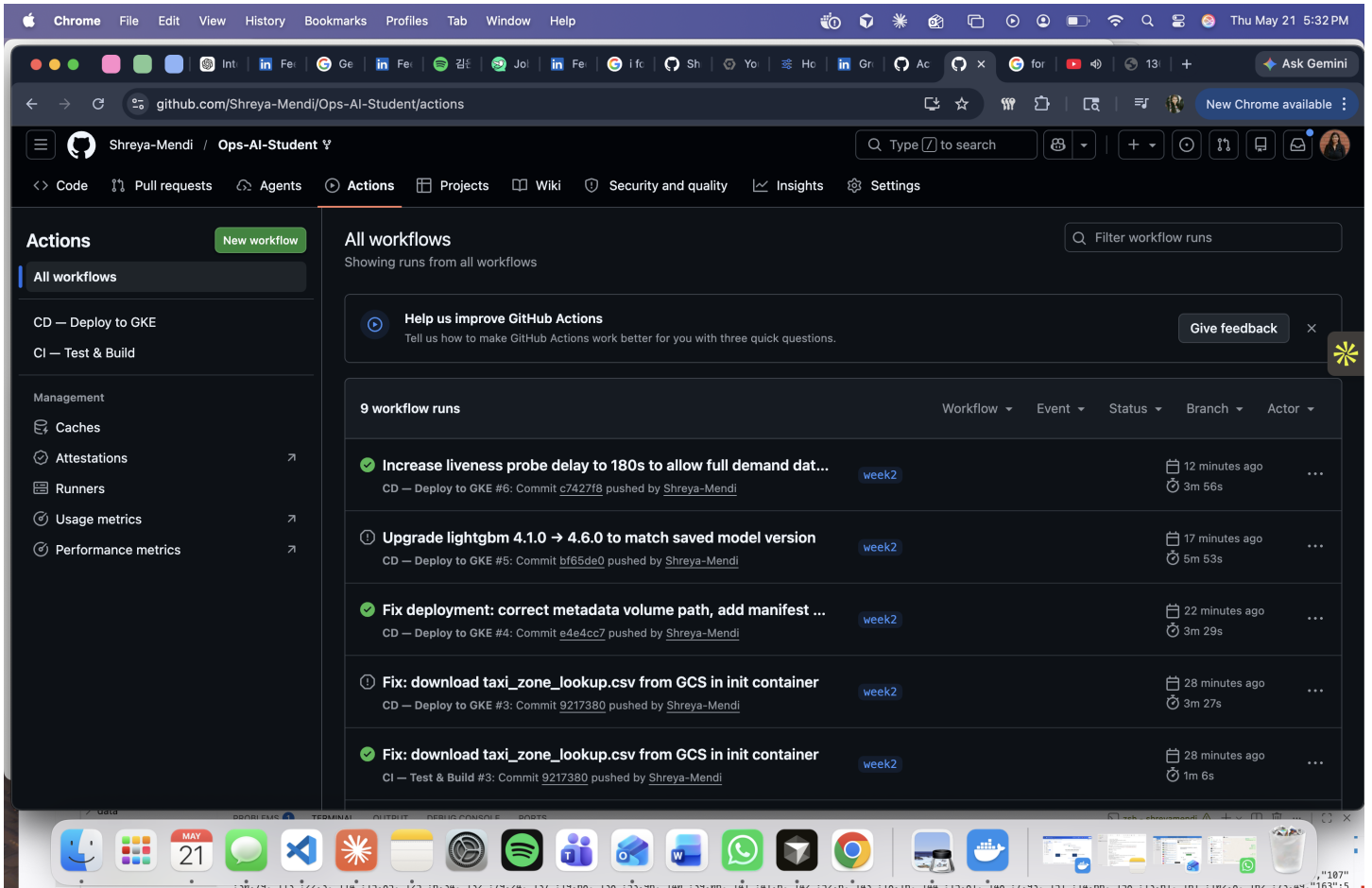
Forecast API Response

```
● Bash(curl -s "http://136.112.78.118/api/forecast?zone_id=68&hour=17&dow=4&steps=4&date=2026-05-21" 2>&1)
[ [
  {
    "time_bucket": "2026-05-21T21:40:12.430470",
    "predicted_trips": 6.44,
    "hour": 21,
    "minute": 40
  },
  {
    "time_bucket": "2026-05-21T21:55:12.430470",
    "predicted_trips": 38.75,
    "hour": 21,
    "minute": 55
  },
  {
    "time_bucket": "2026-05-21T22:10:12.430470",
    "predicted_trips": 20.51,
    "hour": 22,
    "minute": 10
  },
  {
    "time_bucket": "2026-05-21T22:25:12.430470",
    "predicted_trips": 32.88,
    "hour": 22,
    "minute": 25
  }
]
```

kubectl get pods (1/1 Running) + LoadBalancer External IP

```
● shreyamendi@Mac-1131 ~ % kubectl get pods && kubectl get svc demand-api
NAME                                READY   STATUS    RESTARTS   AGE
demand-api-7d4558b8d6-c9dgp        1/1     Running   0           10m
demand-api-7d4558b8d6-wk6fh        1/1     Running   0           10m
NAME                                TYPE           CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
demand-api                         LoadBalancer  34.118.228.252   136.112.78.118   80:30829/TCP     63m
○ shreyamendi@Mac-1131 ~ %
```

GitHub Actions — CD Pipeline Green



GKE Cluster Deletion Confirmation

```
shreyamendi@Mac-1131 ~ % kubectl get pods && kubectl get svc demand-api
NAME                READY   STATUS    RESTARTS   AGE
demand-api-7d4558b8d6-c9dgp   1/1     Running   0          10m
demand-api-7d4558b8d6-wk6fh   1/1     Running   0          10m
NAME                TYPE          CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
demand-api   LoadBalancer  34.118.228.252   136.112.78.118   80:30829/TCP     63m
shreyamendi@Mac-1131 ~ % gcloud container clusters delete operationalizing-ai --zone us-central1-a --project ops-ai-shreya
The following clusters will be deleted.
- [operationalizing-ai] in [us-central1-a]

Do you want to continue (Y/n)? y

Deleting cluster operationalizing-ai...done.
Deleted [https://container.googleapis.com/v1/projects/ops-ai-shreya/zones/us-central1-a/clusters/operationalizing-ai].
shreyamendi@Mac-1131 ~ %
```